

Venkateswara Reddy Satti

Machine Learning Engineer | Production ML Systems | GenAI | MLOps

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Professional Summary

Machine Learning Engineer with 4+ years of experience designing production ML systems, real-time inference APIs, distributed data pipelines, and cloud-native AI infrastructure across fraud, risk, GenAI/RAG, and MLOps.

Professional Experience

Machine Learning Engineer, Real-Time ML Systems | Twilio | Atlanta, GA | Jun 2024 - Present

- Built real-time SMS spam and fraud detection systems using anomaly detection, ensemble learning, PyTorch, and Scikit-learn to improve messaging trust and reduce fraudulent traffic.
- Designed event-driven ML pipelines and FastAPI inference services for high-volume streaming data, supporting low-latency model serving in production.
- Developed GenAI and conversational AI workflows using LangChain, RAG, embeddings, vector databases, and LLM orchestration for automated customer interactions.
- Built feature engineering, model monitoring, drift detection, MLflow evaluation, and observability workflows for production model stability.
- Optimized inference services with batching, caching, asynchronous APIs, and microservices, improving throughput by 30%.

Machine Learning Engineer, Fraud & Risk Systems | FIS Global | Bengaluru, India | May 2021 - Dec 2023

- Built real-time transaction fraud detection models using XGBoost, Scikit-learn, anomaly detection, and ensemble methods, improving fraud detection accuracy by 20%.
- Developed AML detection systems with graph-based anomaly detection, behavioral analytics, and sequence modeling to identify suspicious transaction behavior.
- Built distributed ETL, SQL-based feature pipelines, credit risk models, and FastAPI fraud scoring services, reducing scoring latency by 25%.
- Implemented SHAP explainability, cross-validation, retraining, drift monitoring, and model governance workflows for regulated financial systems.

Selected Projects

Real-Time Fraud Scoring Platform: Designed streaming features, fraud scoring APIs, anomaly detection, explainability, and monitoring for high-throughput transaction systems.

RAG-Based Conversational AI: Built FastAPI applications with LangChain, vector embeddings, retrieval pipelines, prompt engineering, and scalable LLM inference.

Computer Vision Segmentation Systems: Implemented object detection and instance segmentation with Mask R-CNN, DETR, CLIP, and PyTorch.

MLOps Deployment Framework: Created Dockerized pipelines with GitHub Actions, MLflow, model serving, monitoring, and cloud deployment workflows.

Technical Skills

Languages	Python, SQL, Java, Bash/Shell Scripting
ML	PyTorch, TensorFlow, Scikit-learn, XGBoost, Deep Learning, Reinforcement Learning, Anomaly Detection
GenAI	LangChain, RAG, Vector Databases, Embeddings, Prompt Engineering, Conversational AI, LLM Inference
MLOps	MLflow, CI/CD, Model Deployment, Drift Detection, Observability, Experiment Tracking, Model Serving
Backend/Data	FastAPI, REST APIs, AsyncIO, Microservices, ETL, Feature Engineering, Batch & Streaming Pipelines
Cloud/Tools	AWS, Azure, GCP, Docker, Kubernetes, GitHub Actions, Linux, Jupyter, Postman

Education

Saint Louis University | M.S. in Artificial Intelligence, GPA 3.8/4.0 | St. Louis, MO | 2024 - 2025

Coursework: Machine Learning, Deep Learning, Computer Vision, Data Mining, Statistics, Distributed Systems

Certifications

Oracle Generative AI Professional | DRDO Certified Artificial Intelligence Engineer